MILITARY SPECIFICATION SHEET

SEMICONDUCTOR DEVICE, TRANSISTOR, PNP, GERMANIUM, ALLOY-JUNCTION

TYPE 2N123

1. SCOPE

1.1 Scope. This specification covers the detail requirements for a PNP Germanium Alloy-Junction Transistor, medium speed switching, and is in accordance with MIL-S-19500 except as otherwise stated herein.

1.2 Physical dimensions. See figure 1.

1.3 Maximum ratings:

P_T 1/	V_{CB}	V_{EB}	V_{CE}	T_J	T_{stg}
mw	Vdc	Vdc	Vdc	°C	°C
150	-20	-10	-15	85	-65 to +85

1/ Derate 2.5 mw/°C for T_A 25°C.

1.4 Primary electrical characteristics.

	h_{FE}	h_{fe}	V_{CE}(sat)	C_{obo}	f_{hfb}
	V_{CE} = -1 V I_C = -10 ma	V_{CE} = -5 Vdc I_E = -1 ma f = 270 Hz	I_C = -10 ma I_B = -0.5 ma	V_{CB} = -5 Vdc I_C = -1 ma f = 1 MHz	V_{CB} = -5 V I_E = -1 ma
			Vdc	pf	MHz
MIN	30	60	----	--	5
MAX	150	110	-0.2	25	---

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1.4 Primary electrical characteristics. (Cont'd)

	SWITCHING TIME			
	t_d	t_r	t_s	t_f
	n sec	n sec	n sec	n sec
MIN	---	---	----	---
MAX	200	500	1000	400

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-S-19500 Semiconductor Devices, General Specification for.

STANDARDS

MILITARY

MIL-STD-202 Test Methods for Electronic and Electrical Component Parts.

MIL-STD-750 Test Method for Semiconductor Devices.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General. Requirements shall be in accordance with MIL-S-19500, and as specified herein.

3.2 Abbreviations, symbols, and definitions. The abbreviations, symbols, and definitions used herein are defined in MIL-S-19500.

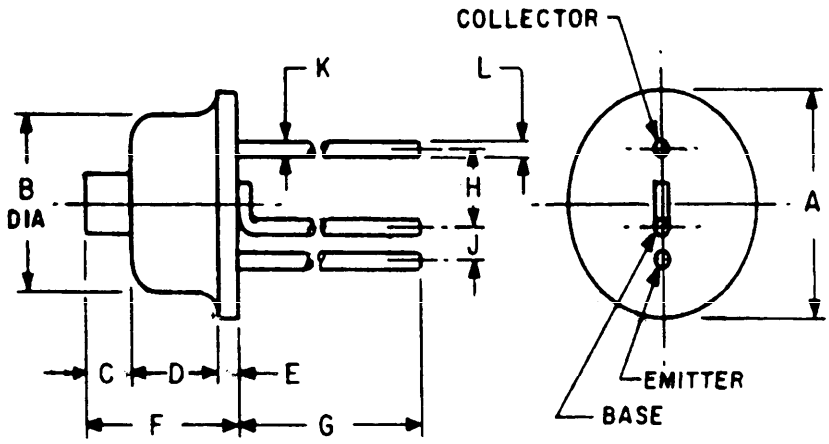
3.3 Design and construction. Transistor shall be of the design, construction, and physical dimensions shown on figure 1.

3.4 Performance characteristics. Performance characteristics shall be as specified in tables I, and II.

3.5 Marking. The following marking specified in MIL-S-19500 may be omitted from the body of the transistor at the option of the manufacturer:

a. Country of origin
b. Manufacturer's identification

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DIMENSIONS

LTR	INCHES		MM		NOTES	LTR	INCHES		MM		NOTES
	MIN	MAX	MIN	MAX			MIN	MAX	MIN	MAX	
A	.430	.460	10.92	11.68		G	1.5		38.10		
B	.322	.345	8.18	8.76		H	.137	.151	3.48	3.84	
C		.135		3.43		J	.041	.055	1.04	1.40	
D	.165	.195	4.19	4.95		K	.016	.020	.41	.51	2,3
E	.020	.045	.51	1.14		L	.016	.026	.41	.66	4
F		.375		9.53							

NOTES:

1. METRIC EQUIVALENTS (TO THE NEAREST .01 MM) ARE GIVEN FOR GENERAL INFORMATION ONLY AND ARE BASED UPON 1 INCH = 25.4 MM
2. UNCONTROLLED FOR .080 INCH (2.03 MM) BELOW SEATING PLANE
3. LEAD DIAMETER FROM .080 INCH (2.03 MM) TO .250 INCH (6.35 MM) BELOW SEATING PLANE
4. LEAD DIAMETER FROM .250 INCH (6.35 MM) BELOW SEATING PLANE TO FULL LENGTH
5. BASE LEAD ELECTRICALLY CONNECTED TO TRANSISTOR SHELL

FIGURE 1. Outline dimensions Transistor type JAN 2N123

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4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection shall be in accordance with MIL-S-19500, and as specified herein.

4.2 Preproduction inspection. Preproduction inspection shall consist of the examinations and tests specified in Tables I and II.

4.3 Quality conformance inspection. Quality conformance inspection shall consist of group A and B inspections. When specified in the contract or order, one copy of the quality conformance inspection data, pertaining to the device inspection lot, shall be supplied with each shipment by the device manufacturer.

4.3.1 Group A inspection. Group A inspection shall consist of the examinations and tests specified in Table I.

4.3.2 Group B inspection. Group B inspection shall consist of the examinations and tests specified in Table II.

4.4 Methods of examination and test. Methods of examination and test shall be as specified in Tables I and II.

4.4.1 Pulse measurements. Conditions for pulse measurement shall be as specified in Section 4 of MIL-STD-750.

4.4.2 Resistance to solvents. Transistors shall be subjected to tests in accordance with Method 215 of MIL-STD-202. The following details shall apply:

a. All areas of the transistor body where marking has been applied shall be brushed.

b. After subjection to the tests, there shall be no evidence of mechanical damage to the device, and markings shall have remained legible.

4.4.3 Hermetic seal tests.

4.4.3.1 Fine leak tests. All devices shall be fine-leak tested in accordance with MIL-STD-750, Method 1071.

4.4.3.2 Gross leak tests. All devices shall be tested for gross leaks in accordance with MIL-STD-750, Method 1071.

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TABLE I. Group A Inspection

Examination or test	MIL-STD-750		LTPD	Symbol	Limits		
	Method	Details			Min	Max	Unit
Subgroup 1			10				
Visual and mechanical examination	2071	See Figure 1		---	---	----	----
Subgroup 2			5				
Breakdown voltage, collector to emitter	3011	Bias Cond. D I_C = -600 ua		BV_{CEO}	-15	----	Vdc
Breakdown voltage, collector to base	3001	Bias Cond. D I_C = -600 ua		BV_{CBO}	-20	----	Vdc
Collector to base cutoff current	3036	Bias Cond. D V_{CB} = -20 Vdc I_E = 0		I_{CBO}	---	-6	uAdc
Emitter to base cutoff current	3061	Bias cond. D V_{EB} = -10 Vdc I_C = 0		I_{EBO}	---	-6	uAdc
Forward-current transfer ratio	3076	V_{CE} = -1 Vdc I_C = -10 ma Pulsed (see 4.4.1)		h_{FE}	30	150	---
Small signal short-circuit forward-current transfer ratio	3206	V_{CE} = -5 Vdc I_E = -1 ma f = 270 Hz		h_{fe}	60	110	---
Collector to emitter voltage (saturated)	3071	I_C = -10 ma I_B = -0.5 ma		$V_{CE}(sat)$	---	-0.2	Vdc
Switching characteristics		See figure 2					
Delay time	---	I_C = -10 ma		t_d	--	200	n sec
Rise time	---	I_{b_1} = I_{b_2} = 1 ma		t_r	--	500	n sec
Storage time	---			t_s	--	1000	n sec
Fall time	---			t_f	--	400	n sec

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TABLE I. Group A Inspection - Continued

Examination or test	MIL-STD-750		LTPD	Symbol	Limits		
	Method	Details			Min	Max	Unit
Subgroup 2 - Cont'd							
Open circuit output capacitance	3236	I_C = -1 ma V_{CB} = -5 Vdc f = 1 MHz		C_{obo}	---	25	uuf
Small-signal, short-circuit, forward-current, transfer-ratio, cutoff frequency (common base)	3301	V_{CB} = -5 Vdc I_E = -1 mA		f_{hfb}	5	---	MHz

TABLE II. Group B Inspection

Examination or test	MIL-STD-750		LTPD	Symbol	Limits		
	Method	Details			Min	Max	Unit
Subgroup 1			20				
Physical Dimension	2066	(See figure 1)		---	--	---	---
Subgroup 2			15				
Solderability	2026			---	--	---	---
Thermal shock (temperature cycling)	1051	Test Cond. A, except minimum temperature will be -65°C		---	--	---	---
Thermal shock (glass strain)	1056	Test Cond. A, except max. temperature will be 85°C instead of 100°C		---	--	---	---
Hermetic seal	1071	Test Cond. A, C, D, or F for gross leaks Test Cond. G or H for fine leaks.		---	--	$1X10^{-7}$	atm cc/sec
Moisture resistance	1021			---	--	---	---

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TABLE II. Group B Inspection - Continued

Examination or test	MIL-STD-750		LTPD	Symbol	Limits		
	Method	Details			Min	Max	Unit
Subgroup 2 - Cont'd							
End points:							
Collector to base cutoff current	3036	Bias Cond. D V_{CB} = -20 Vdc		I_{CBO}	--	-6	uAdc
Emitter to base cutoff current	3061	Bias Cond. D V = -10 Vdc		I_{EBO}	--	-6	uAdc
Forward-current transfer ratio	3076	I_C = -10mAdc V_{CE} = -1 Vdc		h_{FE}	30	150	---
Subgroup 3			15				
Shock	2016	Nonoperating; 1500G, approx. 0.5 msec, 5 blows in each orientation: X_1, Y_1, Y_2, and Z_1		---	--	---	---
Vibration, variable frequency	2056	Nonoperating		---	--	---	---
Constant acceleration	2006	10,000 G; in each orientation: X_1, Y_1, Y_2, and Z_1		---	--	---	---
End points:							
(same as subgroup 2)							
Subgroup 4			15				
Terminal strength (lead fatigue)	2036	Test Cond. E		---	--	---	---
Subgroup 5							
Salt atmosphere (corrosion)	1041			---	--	---	---

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TABLE II. Group B Inspection - Continued

Examination or test	MIL-STD-750		LTPD	Symbol	Limits		
	Method	Details			Min	Max	Unit
Subgroup 6			15				
Resistance to solvents	----	MIL-STD-202, Method 215 (see 4.4.2)		---	--	---	---
Subgroup 7			5				
High-temperature life (nonoperating)	1031	T_{stg} = +85°C t = 340 hrs					
End points:							
Collector to base cutoff current	3036	V_{CB} = -20 Vdc I_E = 0		I_{CBO}	---	-6	uAdc
Forward current transfer ratio	3076	V_{CE} = -1 Vdc I_C = -10 ma		h_{FE}	30	150	
Subgroup 8			λ=7				
Steady state operation life	1026	P_T = 150 mw V_{CB} = -10 Vdc t = 1000 hrs					
End points:							
(same as subgroup 7)							

5. PREPARATION FOR DELIVERY

5.1 Preparation for delivery and the quality assurance provisions for preparation for delivery shall conform to MIL-S-19500.

6. NOTES

6.1 Notes. The notes specified in MIL-S-19500 are applicable to this specification.

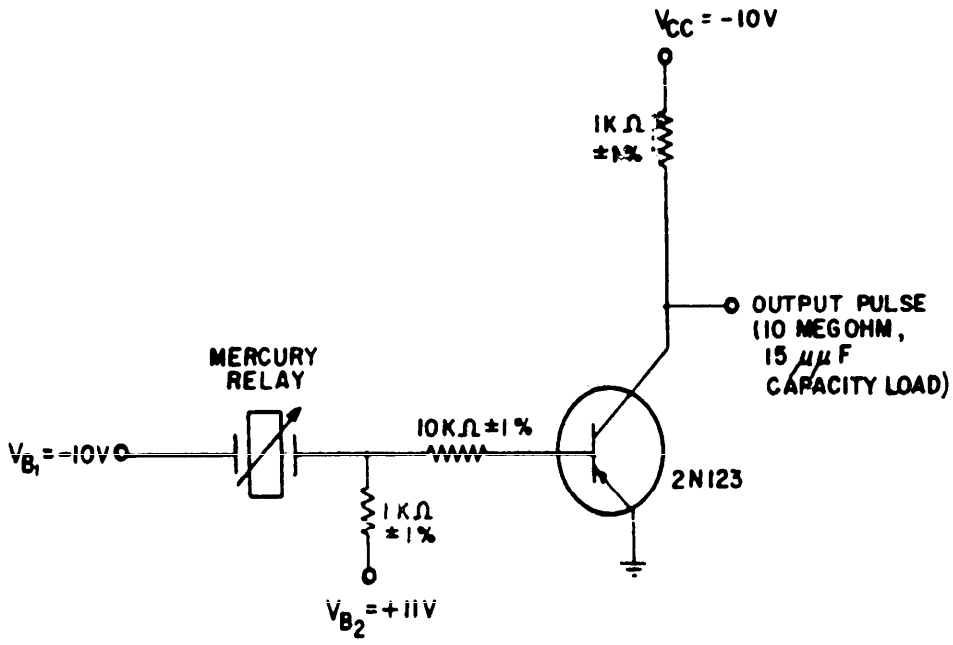
Custodian:
Air Force - 85

Preparing activity:
Air Force - 85

(Project 5961-F253)

Review activities:
Air Force - 11, 17, 70

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I_C = -10 mA

$I_{B_1} = I_{B_2}$ = 1 mA

h_{FE} FORCED TO 10

FIGURE 2. Switching Circuit

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